Personalized Visiting Route Generation for Virtual Museums Based on Large-Scale Models

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Abstract. Virtual museums, as key platforms for cultural dissemination, struggle with non-personalized visiting routes and limited immersive experiences. Large-scale models, with their natural language processing and image understanding capabilities, offer solutions by enabling intelligent Q&A and image analysis. This paper presents a personalized visiting route generation system for virtual museums based on large-scale models. The system optimizes routes through three main components: (1) sampling point generation and scoring using LVLMs; (2) route generation and smoothing based on user interests; and (3) dynamic route optimization via user feedback and LLMs. It leverages the traveling salesman problem and A* algorithm for efficient route planning, and enhances user interaction through voice input and virtual reality visualization. This paper also designs user experiments to qualitatively and quantitatively demonstrate the effectiveness and convenience of the method.Results show significant improvements in user satisfaction and personalized experience.

Keywords: Large-Scale Models· Human-computer interaction· Virtual reality · Route generation.

1 Introduction

With the rapid development of VR and AR technologies, virtual museums have emerged as a novel form of cultural dissemination. However, most visitors are still confined to fixed touring routes, which undermines immersion. The lack of personalization and automation in the touring experience further diminishes user engagement.

We innovatively apply large-scale models to the generation and optimization of virtual museum tour routes, focusing on developing an intelligent, dynamic, and interactive virtual museum tour route generation system.The system first analyzes the scene, then leverages the natural language processing capabilities of large language models(LLMs) to understand and process user intentions. Specifically, our contributions are threefold:

- A novel pipeline for virtual museum tour route generation based on large-scale models.

- Sampling point generation and scoring using large visual-language models(LVLMs).
- Dynamic route optimization via user feedback and large language models(LLMs).

2 Related Work

2.1 Scene Navigation

Scene navigation technology aids efficient target locating. Wang Lili et al. propose a method to automatically determine the number and location of portals and reduce back-tracing[18]. Wang Miao et al. merge isolated rooms into an integrated environment, allowing users to actively shuttle between different rooms[19]. Goal-driven visual navigation is also a core focus.Zhu Yuke et al. apply deep RL for generalizable indoor navigation, which achieves generalization capabilities for new targets and scenes[24].Hongcheng Wang et al. propose the Demand-Driven Navigation (DDN) task, which aims to find objects without the exact names.Wang et al. propose a method to extract and align textual attribute characteristics with visual attribute features using the CLIP model to enhance scene perception during navigation[17].

2.2 Camera Path Planning

The quality of path planning algorithms is key to determining the user navigation experience. Early classic path planning algorithms include the Dijkstra algorithm[5], A* algorithm[6], and RRT algorithm[9]. With the advancement of neural networks, they have been increasingly integrated into path planning research.Chen et al. proposed a topology planning method based on the Transformer, which integrates visual and language information to generate better camera paths[4]. Some studies focus on generating camera trajectories that better align with human intentions and aesthetic standards. Li et al. proposed the Director3D framework, which uses a Transformer model to model the distribution of camera trajectories and can effectively generate camera trajectories and 3D scenes in the real world[10]. Jiang Hongda et al. addressed the problem of camera trajectory planning in virtual cinematography by using example videos to control the desired shooting methods and achieved camera trajectory prediction in new scenes[8].

2.3 Large Models

Large Language Models From 2019 to 2023, GPT-2/3/4 demonstrated the generative and generalization power of LLMs[13,2,12].Subsequent works—LLMR, SonifyAR, DynaVis and UFO extend the same paradigm to 3D scenes, sound, visual editing, and OS control[15,14,16,23].Li's SituationAdapt further injects visual context, enabling real-time multimodal optimization of MR layouts and showcasing the potential of unified large-model reasoning in immersive environments[11].

3 Methodology

3.1 Overview

We design a personalized tour route generation system, leveraging the natural language processing and image understanding capabilities of large-scale models to analyze scenes and automatically generate tour routes for users. As shown in Fig. 1, it first analyzes the scene and selects camera poses to capture key images. LVLMs then score these images to create an initial route. After the user chooses a starting point, a personal path is instantiated. During the visit, voice feedback on needs is processed, enabling the system to dynamically adjust the route.

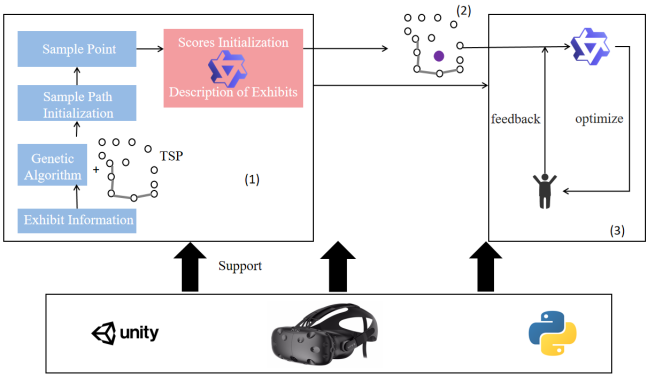


Fig. 1. System Architecture.(1) Sampling point generation and scoring using large visual-language models.(2) Route generation and smoothing based on user interests.and (3) Dynamic route optimization via user feedback and large language models.

3.2 Sampling Point Generation and Scoring Using LVLMs

Sampling Path Initialization Based on Scene Analysis As shown in Fig. 2 and Fig. 3, invalid sampling points are those from which no exhibits are visible, while valid sampling points are those that clearly show exhibits. The higher the proportion of valid sampling points, the higher the degree of data effectiveness processed by the system. As shown in Fig. 5, a View includes an angle(camera rotation), and a score, which represents the score of the image obtained by sampling at this point. Each sampling point includes a position, a target exhibit and three Views.Thus, scoring images becomes the problem of how to construct sampling points. The algorithm defines the R-point of an exhibit as a special sampling point. Its position is obtained by displacing the exhibit's mesh center along its forward vector:

$$Position = Target_{meshCenter} + Target_{forward} * s \tag{1}$$

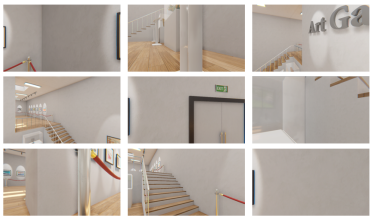


Fig. 2. Invalid Sampling Points

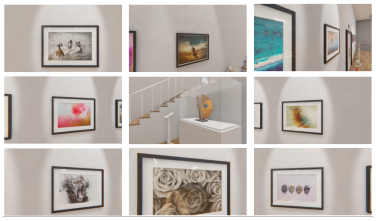


Fig. 3. Valid Sampling Points

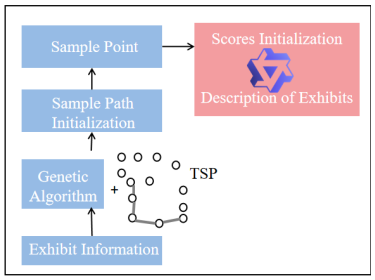


Fig. 4. Sampling Point Generation and Scoring Using LVLMs

$$s = 0.8 * ActualSize/0.6336 \tag{2}$$

The constants 0.8 and 0.6336 were determined by manually tuning s for each object in our sample scene and then fitting the resulting optimal distances to the objects' actual sizes.

We formulate the exhibit-visiting-order problem as a Traveling Salesman Problem and employ a heuristic genetic algorithm to solve it[22]. The entire scene is partitioned into multiple square grids($MapGrid$). The size of each $MapGrid$ is $0.3\,\mathrm{m} \times 0.3\,\mathrm{m}$(according to the average object size in the sample scene). These $MapGrids$ are used as nodes in the A* algorithm to generate a rough path using $orders$[6].The final output is a $MapGrid_path$ as illustrated in Fig. 6.

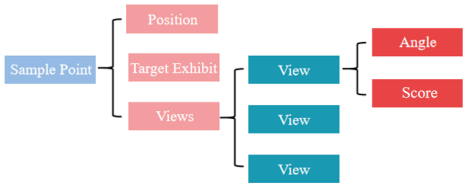


Fig. 5. Sampling Point Structure.

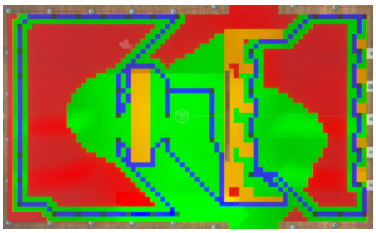


Fig. 6. Generated Rough Path

Sampling Point Generation The sampling area is defined as a collection of *MapGrid* centered on the *MapGrid* that contains the R-point, forming a square of size $(2R+1) \times (2R+1)$. This area is constructed dynamically along the coarse path produced by the A* algorithm and the set of R-points. From each area we generate k sampling points: one central point (coinciding with the R-point) and two additional points drawn uniformly at random within the area. Empirical tests show that $k = 3$ minimizes the total number of sampling points while still guaranteeing that every exhibit is captured. The central sampling point is determined by the position of the R-point, while the two random sampling points are randomly generated within the sampling area. The camera at each sampling point looks along the vector $\mathbf{v}_1$ from the point to its assigned exhibit. Through the entire algorithm, the effective rate of sampling points can reach 100% for every scene.

Scoring Initialization Based on LVLMs During the scoring initialization phase, the LVLMs need to complete two tasks: (1) exhibit-content understanding, carried out by the Exhibit-Understanding Agent, and (2) viewpoint scoring, carried out by the Scoring Agent. The system-level prompt provided to the exhibit understanding Agent instructs the LVLMs to focus on the type, appearance, color, and content of the exhibit, and to respond in a fixed format. In the second task, Sampling points with low scores and unsuitable viewpoints can be preliminarily filtered out.

3.3 Route Generation and Smoothing Based on User Interest

After obtaining the scores, users can freely select a starting point I. The system locates the nearest sampling point P and designates its associated sampling area as the initial area A_0. Starting from A_0, the system visits all sampling areas in sequence according to *orders* $(A_1, \ldots, A_n)$, generating the tour path. First, the camera performs a smooth transition from I to P_0. Within each area A_i the point with the highest score is selected as the key sampling point P_i. The ordered sequence $\{P_i\}_{i=0}^{n}$ thus constitutes the critical way-points of the final path. Since A* plans by cost, we label colliders (walls, display cases, pillars) as occupied and others as free.The transition cost from any free cell to an occupied cell is set to ∞, thus guaranteeing obstacle avoidance. Using P_i as the starting point and P_{i+1} as the endpoint, the system successively generates a smooth path using cubic spline interpolation algorithm. The algorithm controls the overall speed based on the average score of all sampling points: the higher the average score, the slower the movement speed during the tour.

3.4 Dynamic Route Optimization via User Feedback and LLMs

As shown in the workflow in Fig. 7, the Intent-Reasoning Agent(IR Agent) first identifies the user's intentions. The IR Agent categorizes user feedback into different types (preference adjustment, viewpoint adjustment, path adjustment,

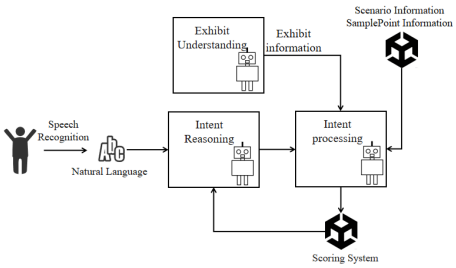


Fig. 7. Dynamic Route Optimization via User Feedback and LLMs.

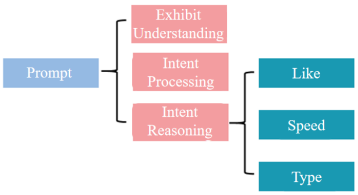


Fig. 8. Prompt Structure

and touring speed). The IR Agent combines the inferred intentions with scene information and sampling point information in an appropriate format and passes them to the Intent Processing Agent(IP Agent). The IP Agent then formulates corresponding processing strategies based on the type of user feedback. As shown in Fig. 8, depending on different tasks, we design four different system-level prompts, corresponding to the Scoring Agent, Exhibit Understanding Agent(EU Agent), IR Agent and IP Agent. During the development of the system, it was found that the LLMs performs poorly with multi-step historical records; overly long historical records can cause the model to forget the initial task and produce unpredictable responses. we therefore preset core prompts for various questions, abandoning historical records each time a question is asked, and using the core prompt combined with the current question for inquiry. Taking the IR prompt as an example, every prompt we design follows two scientific principles: Clarity: instructions must be explicit, specific, and unambiguous; Structure: titles, lists and delimiters are used to organize the prompt so the model can parse complex commands more easily and return equally structured, logical outputs [20]. As

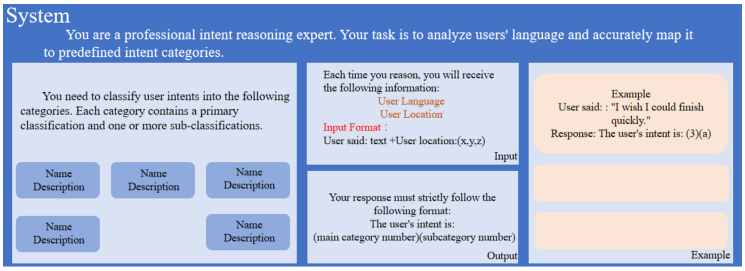


Fig. 9. Prompt Structure.

shown in Fig. 9, we first apply the "persona pattern" by assigning the model an expert identity, establishing a clear framework for its subsequent reasoning[21]. Next, we explicitly enumerate all intent categories. We then specify the required

input and output formats. Finally, following Brown's few-shot prompting principle, we provide a small set of concrete input-output examples to let the model learn by imitation and improve the performance[3].

4 User Experiment

4.1 Experimental Design and Content

Participants and Environment A total of 32 volunteers aged 18 to 24 with basic computer skills were recruited; seven had prior experience with VR applications. The experimental scene was a two-storey, richly furnished virtual museum displaying ancient paintings, abstract art, photography and sculpture. The LVLM used in the experiment is Qwen-VL-Max, and the LLM is Qwen-QwQ.

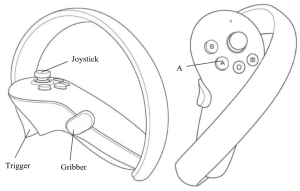


Fig. 10. Controller Button Layout

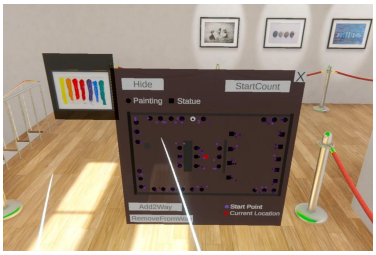


Fig. 11. Comparison Method Operation 1

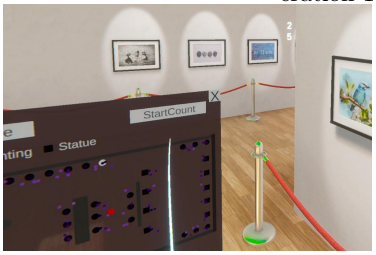


Fig. 12. Comparison Method Operation Two

Experimental Tasks and Procedures The participants first described the exhibits and then used two different methods to locate the exhibits and generate routes, repeating the process for three rounds each.

The experimental methods included a comparison of the conventional ray-casting method (the comparison method) and our method. The comparison method simply lets users click on icons that represent exhibits to see the corresponding images.As shown in Fig. 11 and Fig. 12, users clicked on exhibit

locations on the virtual map to view thumbnails and selected points on the route based on their descriptions, with the order of the generated route determined by the click sequence. In our method, after pressing the left trigger shown in Fig. 10, the user simply speaks the desired exhibit type; the system then generates a route. If the user doubts the route's reasonableness or coverage, they can give additional voice feedback and the system will dynamically adjust the path.

4.2 Result And Discussion

Table 1. Evaluation Metrics.

Metric	Description
Coverage Rate	The percentage of user-required exhibits covered by the paths generated by both methods.
Error Rate	The percentage of selected exhibits that are not part of the correct answer.
Path Length Ratio	The ratio of the path length generated by the conventional ray-casting method to that generated by our method for each question.
Time	The time cost.

User Load and System Usability The NASA Task Load Index (NASA-TLX) categorizes user workload into six dimensions[7]. The System Usability Scale (SUS) consists of 10 statements about the system, each with a 5-point rating scale (1–5) corresponding to "strongly disagree," "disagree," "neutral," "agree," and "strongly agree". The higher the total score, the higher the usability of the system[1].

Table 2. NASA and SUS Scores.

Method	Average NASA Score↓	Average SUS Score↑
Conventional Ray Casting	54.69±5.80	64.56±8.40
Our Method	40.57±6.45	73.05±8.08
P	7.21×10^{-9}	1.20×10^{-5}

The SUS results indicate that users generally expressed higher satisfaction with this method compared to the alternative method($p = 1.20 \times 10^{-5}$). Specifically, the majority of participants found this method to be more consistent and user-friendly. This consistency and ease of use are primarily reflected in the intuitiveness and logical structure of the system's operations.

Table 3. NASA Scores And T-Test

Method	Mental Demand↓	Physical Demand↓	Temporal Demand↓
Our Method	50.00±11.22	46.88±13.43	41.88±8.87
Conventional Ray Casting	44.06±11.10	65.16±7.13	69.84±9.63
P	0.418	1.59×10^{-8}	5.97×10^{-18}
Method	Performance Level↓	Effort↓	Frustration↓
Our Method	35.47±11.60	39.22±8.72	30.00±10.70
Conventional Ray Casting	48.3±13.83	61.25±8.70	39.53±11.53
P	1.63×10^{-4}	9.55×10^{-5}	0.002

The NASA-TLX results show the proposed method significantly outperformed conventional Ray Casting on six of seven subscales: Physical Demand and Temporal Demand dropped by over 20 points ($p = 1.59 \times 10^{-8}$ and $p = 5.97 \times 10^{-18}$), while Performance ($p = 1.63 \times 10^{-4}$), Effort ($p = 9.55 \times 10^{-5}$) and Frustration (p = 0.002) were also improved, reducing the overall score(p < 0.001). Although Mental Demand was slightly higher than the comparison method, the difference was not significant (p = 0.418),highlighting the advantage of the new method in reducing physical, temporal and affective costs.

Coverage Rate and Error Rate As shown in Table 4, our method achieves a higher coverage rate and a lower error rate than the comparison method. Although the t-tests do not reach significance ($p = 0.061$ and $p = 0.574$), both metrics move in a consistently favorable direction; in particular, the p-value for coverage is close to the 0.05 boundary, suggesting that statistical support could be achieved with a larger sample or reduced variability. Moreover, the proposed method exhibits lower standard deviations, indicating a more stable performance among the participants.

Table 4. Coverage Rate and Error Rate.

Method	Coverage Rate↑	Error Rate↓
Conventional Ray Casting	83.99±22.72	6.67±13.62
Our Method	91.31±12.56	4.9±13.85
P	0.061	0.574

Average Path Length Ratio After organizing all the experimental data and excluding cases where the path length do not differ when only one or two objects are included, the average path length ratio was calculated to be 1.45. The average path length generated by our method is significantly shorter than that of the comparison method, indicating that our method is more effective in path optimization than manual selection. Compared to the user's manual and unordered

selection, our method can generate shorter tour paths, improving the efficiency and reducing unnecessary tour distance.

Time The average time required to generate a route with our method was markedly shorter than that of the comparison method ($p = 1.84 \times 10^{-17}$). The result demonstrates that our method is simple and intuitive, yielding a significant improvement in the users' operation experience. We measured the time from when users pressed the button and spoke to when they received system feedback; the average delay was approximately 18 s. Because we invoked a cloud-based API, a large portion of this latency was attributable to network delay. After deploying the Qwen-QwQ model locally, the response time per user feedback dropped sharply, demonstrating the feasibility of real-time interaction. Nevertheless, to maintain inference quality, we continued to use the cloud API during the experiment.

Table 5. Average Time Spent.

Method	Average Time Spent (s)↓
Conventional Ray Casting	128.63±43.16
Our Method	37.50±9.23
P	1.84×10^{-17}

5 Conclusion

We present a personalized tour route generation system for virtual museums. The system analyses the scene, receives and interprets user feedback during the tour, and dynamically adapts the route to deliver a personalized and immersive experience. Through user experiments, we demonstrate the effectiveness and accuracy of our system compared to traditional methods, as well as its ability to increase user satisfaction and reduce user burden. However, our method still has shortcomings. A new initialization is required for each new scene, which objectively consumes a certain amount of time. Additionally, during the user's interaction with the system, the two-tier response process also incurs time costs. In future work, we will explore more intuitive feedback methods, improve response speed, and enhance user experience.

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